

THE PROBLEM NOBODY SOLVED

Privacy.

We all need it. *We don't get it.*

Our most sensitive conversations are now typed, recorded, and transcribed by software that quietly ships every word to a server we will never see.



Some people are trusted with what no one else can hear – and are bound, by oath or by law, to keep it.

Doctors

Therapists

Journalists

Accountants

Clergy

Financial advisers

HR & social workers

Lawyers.

Yet every digital note-taker ships that room to the cloud.

Attorney–client privilege is the strongest confidentiality in law. A first consultation is the most sensitive hour of all.



THAT IS WHAT SUB ROSA SOLVES

The AI note-taker that never leaves the room.

Sub Rosa records a lawyer's first client consultation and turns it into structured intake notes — *entirely on the phone*. The microphone, the transcription, the reasoning, the storage: all on-device. No internet permission. Nothing to leak.

**It listens**ON-DEVICE ASR**It reasons**ON-DEVICE LLM • NPU**It forgets the network**NO CLOUD, BY CONSTRUCTION

From a spoken consultation → a clean intake memo.

Pick a matter type, hit record, and talk. When you stop, Sub Rosa returns three things — and leads your eye to the one that matters most: **what the client forgot to tell you.**

① FACTS

Each fact paired with the **verbatim quote** from the transcript that backs it — so nothing is invented.

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INTAKE ITEMS NOT COVERED

~40

TOK/S • HEXAGON NPU

0

BYTES SENT • OFFLINE

② MISSING INFORMATION

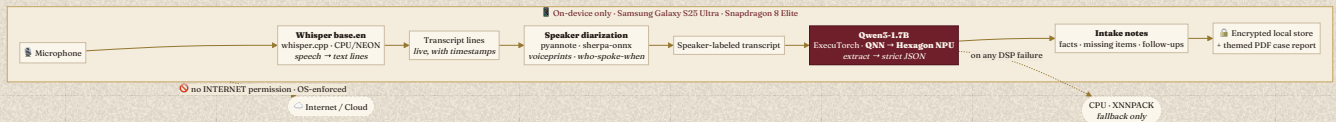
The standard intake items the client **never mentioned** — a true set-difference against a checklist. *The killer feature.*

③ FOLLOW-UP PROMPTS

Short, neutral questions to ask next — never legal conclusions.

► LIVE DEMO *contract-dispute consultation*

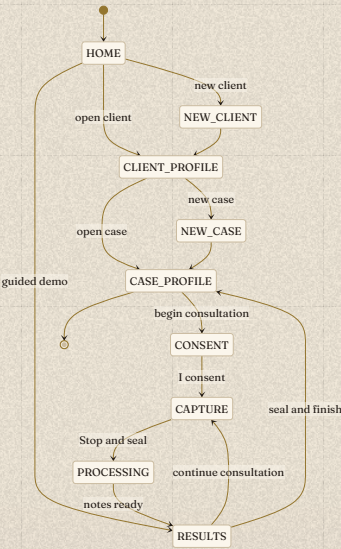
I SYSTEM OVERVIEW – EVERYTHING HAPPENS ON THE PHONE



The LLM – the technical centerpiece – runs on the **Hexagon NPU** via ExecuTorch's QNN backend. CPU/XNNPACK is only the safety net. No INTERNET permission, OS-enforced.

ON SCREEN *the consultation, recording live*

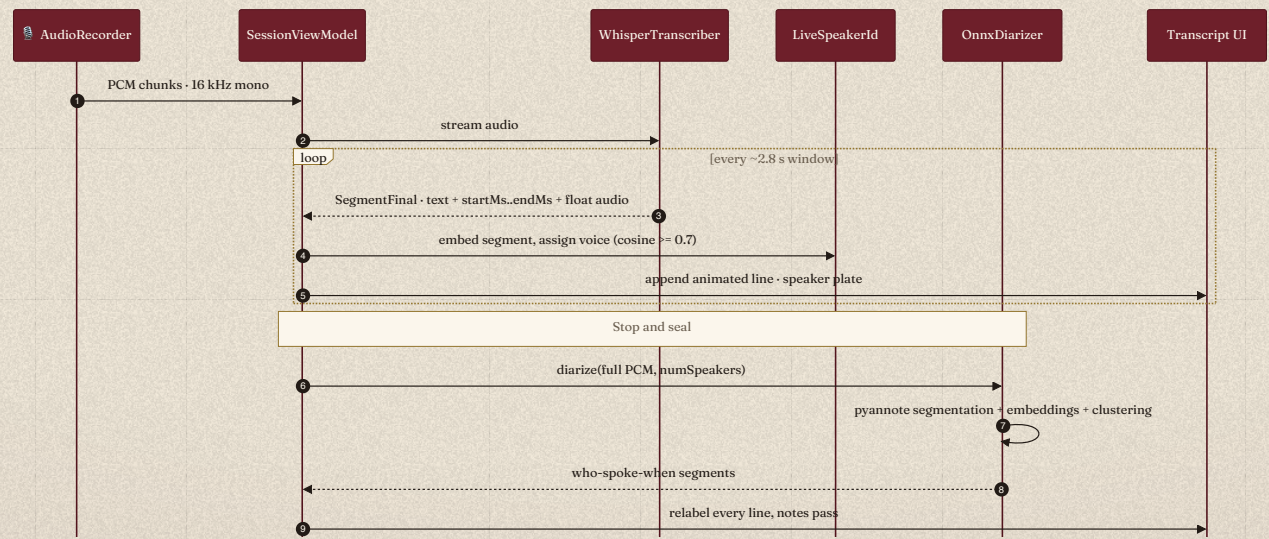
II CONSULTATION FLOW – THE SCREENS, IN ORDER



Client → Case → Consultation. A **case** fixes the matter type, which fixes the intake checklist, and owns its many resumable consultations. MVVM + enum-driven phase navigation.

ON SCREEN *tap through Consent → Capture*

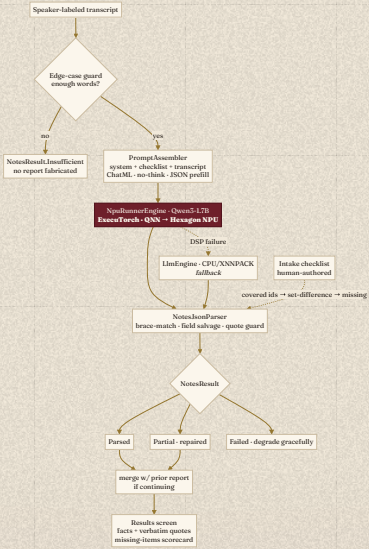
III LIVE CAPTURE & PYANNOTE SPEAKER DIARIZATION



Live voiceprints (`LiveSpeakerId`) split speakers instantly; the full `pyannote` pass (`OnnxDiarizer`) refines at Stop, over sherpa-onnx. The manual tap always overrides.

ON SCREEN *lines appear, speaker plates switch*

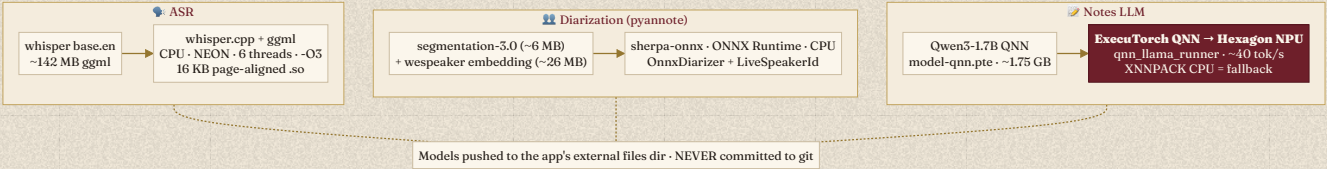
IV NOTES GENERATION - LLM → STRICT JSON



No legal knowledge in the weights. The model only extracts and reformats; missing items are computed in code (checklist - covered). A 1.7B model stays accurate, with zero fine-tuning.

ON SCREEN the missing-info scorecard counts up

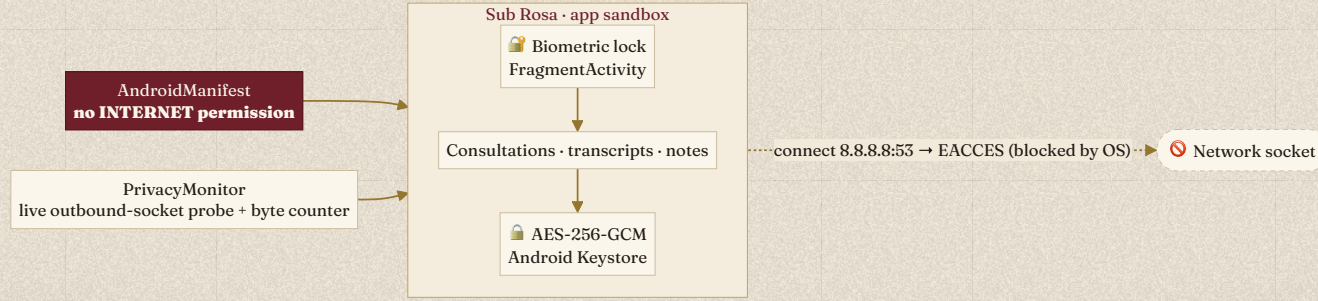
V ON-DEVICE MODELS & RUNTIMES



Notes LLM primary = **Hexagon NPU via QNN** (qnn_llama_runner + model-qnn.ptc, ~40 tok/s); XNNPACK CPU is the fallback.
Models are pushed to the device, never committed to git.

ON SCREEN PROOF panel – backend & tok/s

VI PRIVACY & TRUST – VERIFIABLE ON STAGE



Airplane mode proves nothing a judge can see. Sub Rosa proves it **structurally**: no INTERNET permission, AES-256-GCM at rest, and a live probe (connect 8.8.8.8:53 → EACCES) showing the OS deny the socket.

ON SCREEN bytes 0 - socket blocked

No cloud bills — so no usage rents.

We sell trust, licenses, and content.

RECURRING • B2B

Per-seat license

\$39 / lawyer / month (billed annually). Offline activation keys — no server needed to charge.

RECURRING • B2B

Firm & legal-aid tiers

Volume licensing, MDM / on-prem deployment, and admin controls for practices and clinics.

CONTENT MOAT

Checklist packs

Attorney-reviewed intake packs — immigration, family, PI, IP, criminal — sold as add-ons. The legal value scales here.

LICENSING

White-label engine (SDK)

License the on-device stack to other confidential verticals: **medical intake**, **journalism**, **therapy**. Same architecture.

SERVICES

Professional services

Custom checklist authoring, privilege / HIPAA hardening, deployment, training & support contracts.

WHY IT HOLDS

High margin by design

\$0 server cost per consultation. Value scales with trust and content, not with tokens or API spend.

→ FUTURE WORK

- ◆ **Whisper on the NPU.** Move ASR to the Hexagon via QNN – both stages on the NPU.
- ◆ **Encrypted backup & export.** A portable .subrosa file so a lost device doesn't mean lost notes.
- ◆ **Bigger & multilingual models.** Larger decoders as device RAM allows; non-English intake.
- ◆ **More checklist packs.** A growing, attorney-reviewed library across practice areas.
- ◆ **Long-consultation summarization.** On-device running summary to beat the context window.
- ◆ **Local matter-management & iOS.** Offline-first integrations and a second platform.

Today: the full product loop runs offline on the S25 – Whisper ASR, pyannote diarization, Qwen3-1.7B on the **Hexagon NPU** (~40 tok/s), the missing-info scorecard, encryption, and the live privacy proof.



Thank you.

Questions?

Sub Rosa — on-device AI, under the rose.

github.com/Dileep2896/sub-rosa-executorch